ProofLine

Brand governance that keeps watching. No claim goes live without a source.

Hackathon	CodeStar 30 (We Code Coders), Round 3: final submission
Problem statement	Brand governance, issued by Inkloom AI
Team	UDAYA SANKAR (solo participant)
Submitted by	KAVALI VEERA VENKATA UDAYA SANKAR (team leader, all roles)
Final working link	https://proof-line-sigma.vercel.app
Final repository	https://github.com/udayasankar35/ProofLine
Documents in the repository	README.md, docs/ARCHITECTURE.md, docs/EVALUATION.md, docs/TESTING.md, docs/ROADMAP.md

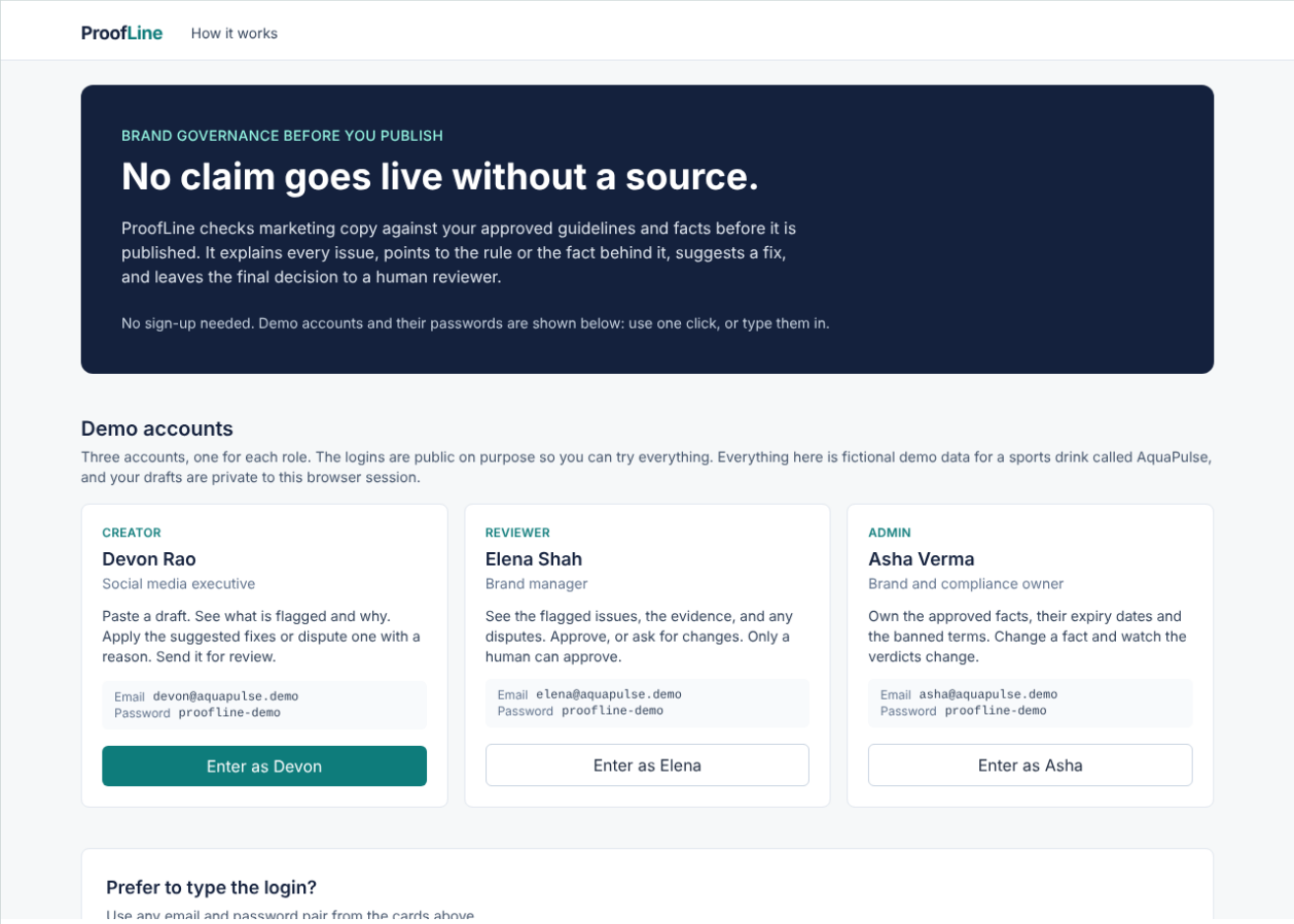
> **In one sentence:** ProofLine checks marketing text, Word files, PDFs and images against a brand's approved guidelines and facts before they are published, explains every finding, and leaves approval to a human. **After approval it keeps watching:** when the evidence behind a claim expires, changes or is removed, the approved piece is re-flagged automatically. **No sign-up.** The demo logins are on the home page.

1. Try it (no help needed)

Role	Email	Password
Creator (Devon Rao)	devon@aquapulse.demo	proofline-demo
Reviewer (Elena Shah)	elena@aquapulse.demo	proofline-demo
Admin (Asha Verma)	asha@aquapulse.demo	proofline-demo

The logins are public on purpose (the organisers suggested it) and protect nothing: all data is fictional and every visitor gets a private workspace. One click on a role card also signs in.

- As **Devon**: **New draft**, load *Instagram caption (risky)*, **Check this draft**. Fix or dispute the findings. Then try the **Upload** tab with the sample tile (image), press release (Word) or flyer (PDF).
- When no high-risk finding is open, **Send to review**. Switch to **Elena** and **Approve**: the claims and the facts behind them are written to the **Evidence ledger**.
- Switch to **Asha**, open **Guidelines & facts** and move a fact's *Valid until* date into the past. The approved piece is **re-flagged** (Review queue, draft page, Ledger). Put the date back and it clears.
- Open **Insights** (use *Load sample history* for a full dashboard) and **Evaluation** (precision and recall for every check, one-click regression run, AI evaluation).



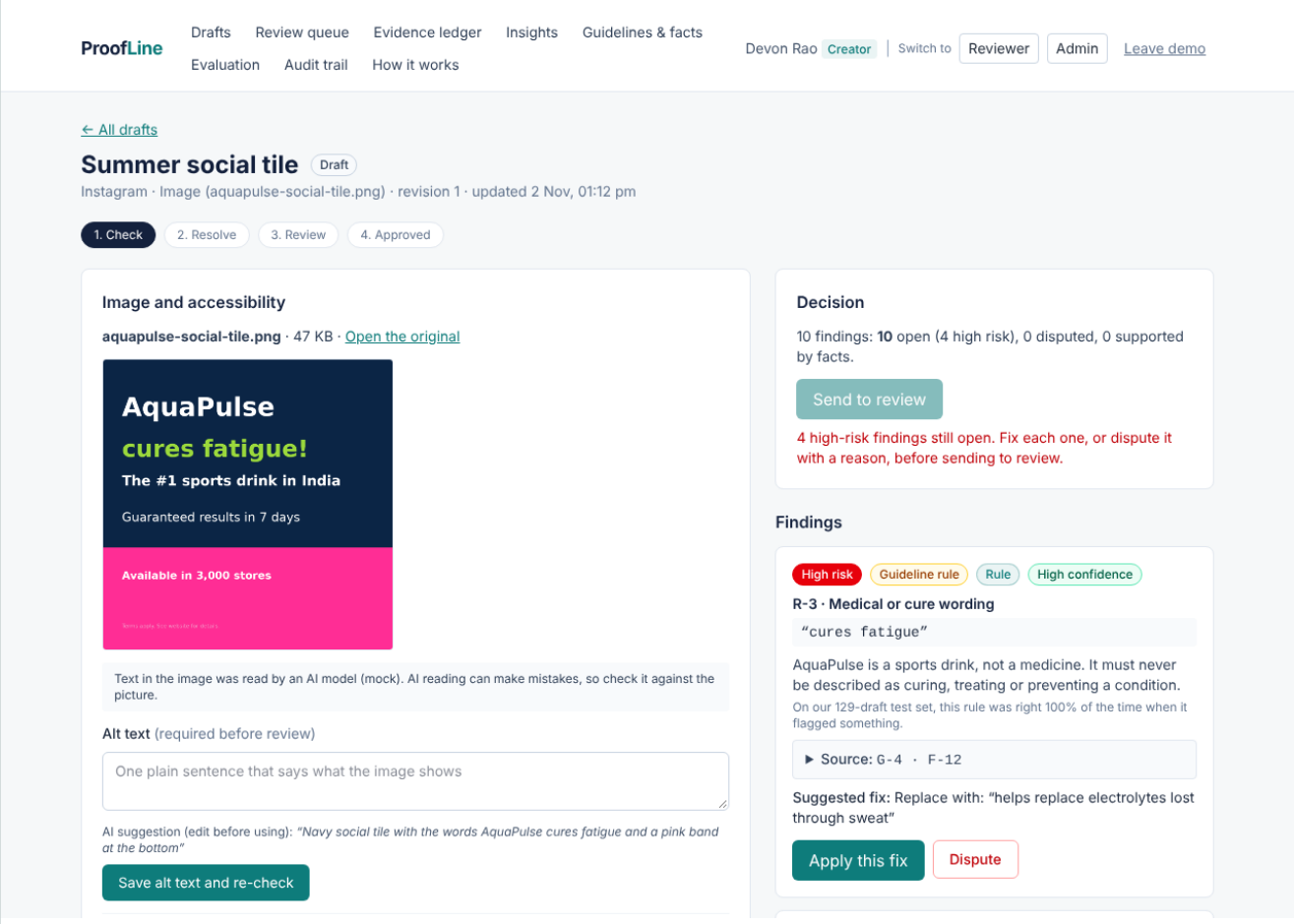
Home page: three demo accounts with their credentials shown, one-click entry, and a typed sign-in form.

2. The three improvements: promised, delivered, and what is still missing

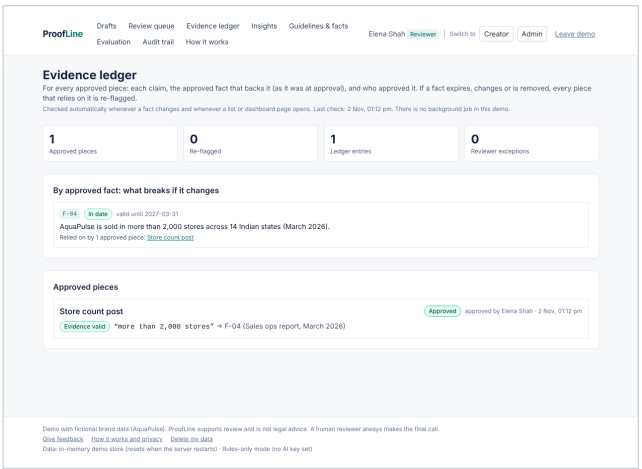
These are the three improvements committed in the Round 2 form. For each one, the table says what was promised, what was built, where to see it, and what is **not** done.

Promised	Delivered	Declared gap
1. Image and document review. Upload DOCX, PDF and images; extract text; run the same checks; image accessibility and visual checks (alt text required before review; brand palette, contrast, small text flagged as AI-assisted).	Word (headers, footers, picture alt text), PDF (text layer directly; scanned PDFs read by a vision model) and PNG/JPEG/WebP (vision model). Text runs through the same rules and claim checks. **R-6** alt text (exact; blocks review), **R-7** low contrast and small text, **R-8** off-palette colours (labelled *AI-assisted, needs human check*, Low confidence). Bad input handled: empty, wrong type, renamed, damaged, password-protected, oversized, no text. Verified on the live site with the real model (section 6).	Fixes change ProofLine's text copy, not the original file. Alt text inside a PDF cannot be verified. Word text boxes, comments and tracked changes are not read. Image estimates are not in the precision/recall set.
2. Evidence ledger and governance insights. Ledger of claims + fact + approver per approved piece; re-flag when a fact expires, changes or is removed; dashboard with most-triggered rules, most-disputed findings, time to approval, facts about to expire.	One ledger line per claim with a snapshot of the fact (statement, source, owner, expiry), the approver and the time; findings approved while open are recorded as reviewer exceptions. Automatic re-flag for **expired, changed and removed** facts, and automatic clearing when the evidence is valid again. Insights page with all four views, an empty state, an *only my drafts* view and optional fictional sample history.	No scheduled background job: the sweep runs after every fact change and whenever a list or dashboard page opens (stated on the page).

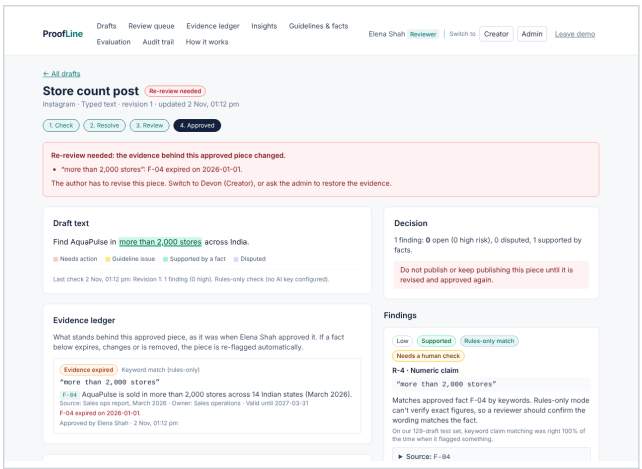
Promised	Delivered	Declared gap
3. Reliability and evaluation suite. 100+ labelled drafts; in-app Evaluation page and README report with precision/recall per rule and for the AI check; one-click regression run; confidence labels on AI findings.	129 labelled drafts (typical, hard, clean controls). **/evaluation** page and **docs/EVALUATION.md** with precision and recall per check, first run versus now, a one-click regression run against a stored baseline, and an AI evaluation that scores the AI step next to keyword matching on the same drafts. Confidence labels (High, Medium, Low) on every finding, checked for whether they mean anything. A regression gate runs before every build.	One person wrote the rules and the labels, so the numbers are optimistic. Images are not in the set. AI numbers are a dated snapshot.



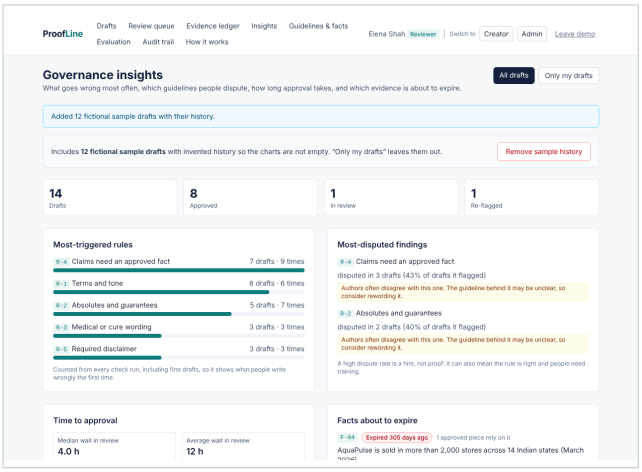
Improvement 1: an uploaded image. The words were read from the picture, alt text is required, and the check shows each finding with its confidence. *Captured from the project's test build, where a stand-in model answers image requests; the live site uses Google Gemini (section 6).*



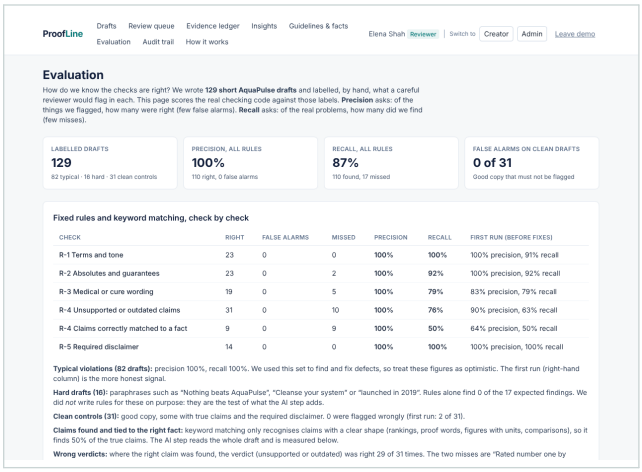
Improvement 2: the evidence ledger after approval. Each claim points to the approved fact behind it and who approved it.



The same piece after an admin let fact F-04 expire: re-flagged for re-review, with the reason on the page.



Improvement 2: governance insights (shown with the optional fictional sample history).



Improvement 3: the Evaluation page. Precision and recall per check, with the first run before fixes beside the current figures.

3. Complete feature list

Area	What it does
Roles and sign-in	Creator, Reviewer, Admin. Public demo logins (email and password) and one-click entry. Server-enforced: only creators edit and submit, only reviewers approve, only admins edit the knowledge base.
Private workspaces	Every browser session has its own drafts, files, ledger and its own copy of the guidelines, facts and terms. Changing a fact never affects another visitor. **Delete my data** purges a session (with a confirmation).
Draft input	Typed text (4,000 characters), or an upload of Word, PDF or an image (4 MB) with sample files for each. Upload a new version, read a file again, download the corrected text.
Fixed rules (no AI)	R-1 banned terms (plural and hyphen variants) and all-caps, R-2 absolutes and guarantees, R-3 medical or cure wording, R-5 required disclaimer (the disclaimer's own words are never flagged), R-6 alt text, R-7 contrast and small text, R-8 palette.
Claims against facts (R-4)	Every factual claim must match an approved, in-date fact. The AI proposes, code verifies (quote must be in the draft, fact must exist, expired fact never supports, figures must match, a comparison needs a fact making the same comparison). Keyword fallback with a visible label.
Explain, fix, dispute	Every finding shows the exact words, reason, source text, a confidence label, how often that kind of finding was right on the test set, and a suggested fix. Fixes create revisions; disputes need a reason and survive re-checks.
Review and approval	Send to review is blocked while high-risk findings are open. Only a reviewer approves; changes require a note. Approved drafts are locked. Double clicks and stale forms are harmless.
Evidence ledger	Claim, fact snapshot, source, owner, approver and time, per approved piece. Reviewer exceptions recorded. Automatic re-flag and clearing.
Governance insights	Most-triggered rules, most-disputed findings, time to approval, facts about to expire (with how many approved pieces rely on each).
Evaluation	129 labelled drafts, precision and recall per check, regression run, AI evaluation, confidence calibration, public JSON at /api/eval.
Audit and reports	Every check, AI request, decision and approval is logged. Downloadable check report. Anonymous tester feedback form.
Knowledge base admin	Guidelines, facts (source, owner, expiry), terms, disclaimer. Add, edit, remove, change expiry, restore the demo set.
Reliability and cost control	25-second AI timeout, backup models, then rules-only with a visible note. AI limits per visitor and per day. Input limits. Friendly errors.

Area	What it does
Accessibility and layout	Labelled fields, visible focus, a skip link; checked for no sideways scroll at 375, 768 and 1280 px.

4. Technical architecture and implementation summary

Layer	Choice	Notes
App	Next.js 16 (App Router, server actions), React 19, TypeScript, Tailwind CSS 4	One application; no separate backend, queue or cron job. All checks and decisions run on the server.
Database	Supabase Postgres, 15 tables, Row Level Security on with no public policies	Reached from the server only with the service key. Knowledge tables are per workspace (composite keys).
File reading	jszip (Word), unpdf (PDF text layer), Gemini vision (images, scanned PDFs)	File type decided from the first bytes, not the name; zip-bomb guard; text clipped at 12,000 characters.
AI	Provider adapter (Gemini default; OpenAI-compatible and Anthropic selectable)	Backup models and a timeout; per-visitor and daily caps; the key stays on the server.
Hosting	Vercel, auto-deploy from the main branch	The build runs the unit and regression tests first.

How a check works

1. Input	2. Read	3. Rules (no AI)	4. Claims and images	5. Findings	6. Human, then watch
Text, Word, PDF or image.	Text from Word and PDF directly; scanned PDFs and images read by the vision model.	Terms, absolutes, medical wording, disclaimer, alt text. Same input, same output.	AI proposes claims; **code verifies** each. Vision estimates contrast, size, palette.	Quote, reason, source, fix, confidence (High, Medium, Low).	Reviewer approves. The ledger records the evidence and re-flags the piece if it breaks.

Why it is dependable. The AI never decides a verdict alone. The evaluation suite calls the *same* code path as the app, so what is measured is what runs. Actions are idempotent, bad input gets a message a person can act on, AI failure falls back to rules, and a test gate blocks any change that makes a check worse than its stored baseline. Full detail is in docs/ARCHITECTURE.md.

5. Improvements completed after Round 2

The three improvements above, and the following changes, were built inside the Round 3 window (31 Oct 23:59 IST to 3 Nov 23:59 IST). The repository history shows each step.

Change	Why
Private per-visitor workspaces extended to the guidelines, facts and terms (Round 2 isolated only drafts)	Admin changes to a fact (needed to try the re-flag) must not affect another visitor testing at the same time.
Idempotent actions; calm notices for double clicks and stale forms	The brief asks for robustness under normal use: same action twice, refresh, back button.
Privacy controls: files kept 3 days, per-session caps, Delete my data (with confirmation), AI disclosure that images and scanned pages are sent to the vision model	Uploaded files are more sensitive than typed text.
Confidence labels and a measured accuracy line on every finding	A reviewer should know where to look hardest.
Demo credentials on the home page and a typed sign-in form	Suggested by the organisers so testers can log in easily.

Change	Why
Tester feedback form and anonymous summary	So real feedback can be collected as soon as anyone uses the product (section 6).
Rewritten About page, README and four documents; sample files for every format	A stranger can clone, run and understand it.
Fixes found by the evaluation: the required disclaimer flagged as a violation; false 'supported' comparisons; plural and hyphen terms; truncated quotes; duplicate findings	Found by the first evaluation run (section 6). All fixed with tests.
Round 2 gaps closed: Word and PDF upload (cut in Round 2) and measured accuracy	Listed as Round 3 work in the Round 2 submission. Still open: testing with real users and CSV import (section 8).

6. Testing details

Layer	What	Result
Unit tests	Rules, fixes, claim verification, AI-reply parsing, visual checks, ledger, Word and PDF extraction (damaged and oversized files), insights, robustness (odd inputs, adversarial text), dataset integrity, regression gate	78 of 78 pass; run before every build
Evaluation set	129 labelled drafts scored for precision and recall (rules-only)	Precision 100%, recall 86.6% (first run before fixes: 93.6% and 81.1%)
End to end (browser)	Core path; upload (image, Word, PDF, bad input); governance (ledger, re-flag on expiry and removal, insights, evaluation, feedback); sign-in; layout at three widths	5 scripts, all pass on a local production build (a stand-in AI model answers the AI paths). The scripts take any web address, so they can be run against the live site.
Live site	Walked through by hand on the live site (real database, real AI) on 3 Nov: demo sign-in as Devon, Elena and Asha; image upload and check; send to review; approve; evidence ledger written (4 lines, 3 reviewer exceptions); fact F-12 set to a past date, which re-flagged the approved piece as 'Re-review needed' with the reason; date restored, which cleared it. Also: /api/health schema check (every table ok), /api/eval, the AI evaluation run (65 drafts, 0 failed requests), and real-model image reading on the sample tile (22 s).	Passed

What the first evaluation run found. Before changing anything we scored the set and kept the result. It found six real defects in the Round 2 engine: (1) the required disclaimer, with its own words 'cure or prevent', was flagged as a violation; (2) 'energy drinks' and 'energy-drink' were missed; (3) **'better than any rival' was marked *supported*** by a fact that only says customers preferred a flavour (the most serious: a false 'safe'); (4) 'India's best-selling' was quoted as 'India's best', so applying the fix would have left '-selling'; (5) 'the leading brand' as a competitor was treated as a ranking claim about us; (6) one fact was reported twice in one sentence. All six are fixed and each has a test. The evaluation then caught a regression of our own while fixing these (a regular-expression slip that cut R-1 recall from 91% to 70%), and a robustness test found that an admin term such as '-' would have matched everywhere. Both were fixed before release.

The AI step, measured.

One live run on 3 Nov 2026 through the deployed app with Google Gemini (gemini-3.5-flash-lite, a backup model), on the 65 drafts that contain a claim, a ranking or a comparison (42 typical, 10 hard, 13 clean controls). 0 requests failed. Figures are **first run, before any change**; cells read precision / recall.

Measure	Keyword matching alone	With the AI step
Claims that are a problem (unsupported or outdated)	100% / 76%	100% / 88% (36 found, 5 missed)
True claims tied to the right approved fact	100% / 50%	100% / 100% (18 of 18)
Right verdict on the problem claims found	29 of 31	34 of 36
Good drafts flagged wrongly (13 clean controls)	0	0

Where it was wrong (7 drafts). It missed '9 out of 10', 'Up to 25%', 'better than', 'Nothing beats' and 'Scientifically formulated', and called '#1' and 'best-selling' *unsupported* where *outdated* was right.

One change was made because of this run. The keyword safety net, which already covered rankings and proof wording, now also covers figures and comparisons the AI did not cover. Re-scored from the same saved answers (no new AI requests), recall rose from 88% to **90% (37 of 41)** with precision still 100%. It was chosen after seeing these failures, so the 90% is optimistic.

Do the confidence labels mean something? Medium findings were right 35 of 35 times and Low findings 13 of 13 times. Both are 100%, so this run does not show that Medium beats Low; the sample is too small. We present the labels as a guide to where a reviewer should look hardest, not as a measured probability. One run, one day, one model; the free AI quota allows two runs a day, so we did not repeat it.

Image reading, tried on the live site with the real model. On the sample tile (text and colours we drew ourselves) Google Gemini read the text, estimated the colours and suggested a usable alt text; we checked each against the file we made. It took 22 seconds, inside the limit. Colour and size values are estimates, which is why every image finding is Low confidence and asks for a human check. This was one trial on one sample, not a measured accuracy.

Tester feedback.

Testing with real users was not done. The Round 2 submission listed testing with 5 creators and 2 reviewers as Round 3 work. We did not recruit outside testers, so there are no human usability numbers here and we do not invent any. What exists instead: an anonymous feedback form (/feedback, summary at /api/feedback) that is live and tested end to end, and holds no responses at the time of submission; and five scripted browser journeys that start from a cold session and use only what is on screen, at phone, tablet and laptop widths. These show that the main path can be finished without prior knowledge of the app. They are not a substitute for people, so real-user testing is the first item on the roadmap.

7. Roadmap

When	What	Why
Next 30 days	**First: test with five creators and two reviewers on their own copy.** Then real accounts and teams (single sign-on); import guidelines and facts from documents; a scheduled background re-check with a notification; Inkloom AI as a provider (the adapter already speaks the OpenAI-compatible API).	Needed before any real brand team can pilot it.
Next 90 days	Evaluate on real, labelled campaign copy with confidence intervals; measure contrast and size from pixels; images in the evaluation set; Word text boxes, comments and tracked changes; apply fixes to the original Word file; Hindi and other Indian languages.	The current numbers are optimistic; images and documents need deeper checks.
Later	Block unapproved content inside publishing tools; learn new terms and facts from reviewer decisions (an admin accepts each); starter policy packs for regulated categories.	From checking to preventing.

8. Limits, responsible design and disclosures

- **Human oversight.** The AI never approves. Only a reviewer approves. Image findings are estimates and are labelled as needing a human check.
- **Not done (declared).** Testing with real users (planned since Round 2). CSV import of guidelines and facts (cut in Round 2, not built in Round 3; roadmap item 2). The domain proofline.udayasankar.xyz is not connected; the public link is the vercel.app address. A scheduled background job was replaced by an on-demand sweep.
- **Honest limits.** Tuned to one made-up brand. Keyword matching only recognises claims with a clear shape; the AI can miss or misjudge a claim. The evaluation set was written by the person who wrote the rules. Fixes edit a text copy, not your original file. No background job. Demo sign-in is not real authentication. ProofLine supports review and is not legal advice.
- **Privacy.** All data is fictional. Each visitor's data is private to their session and deletable at once; uploaded files are kept for 3 days. When AI is on, draft text and the approved-fact list are sent to the AI provider, and for images and scanned PDFs the image or page itself is sent to the vision model. The AI key stays on the server.
- **AI assistance, disclosed.** The code was written with Claude (Anthropic) in a coding session directed, run, tested and deployed by the participant. The product calls a Google Gemini model for claim extraction and image reading. Next.js, React, Tailwind, jszip, unpdf, Supabase and Vercel are the open-source and hosting tools used. All application code for the Round 3 work was written inside the event window; the commit history shows it.
- **Event tools.** As declared in Round 2, the event-provided tools were not used; AI calls use a Google Gemini API key.